Supplementary Figures

Figure S1. Quality assessment of the included studies according to the Cochrane risk-of-bias 2 tool.

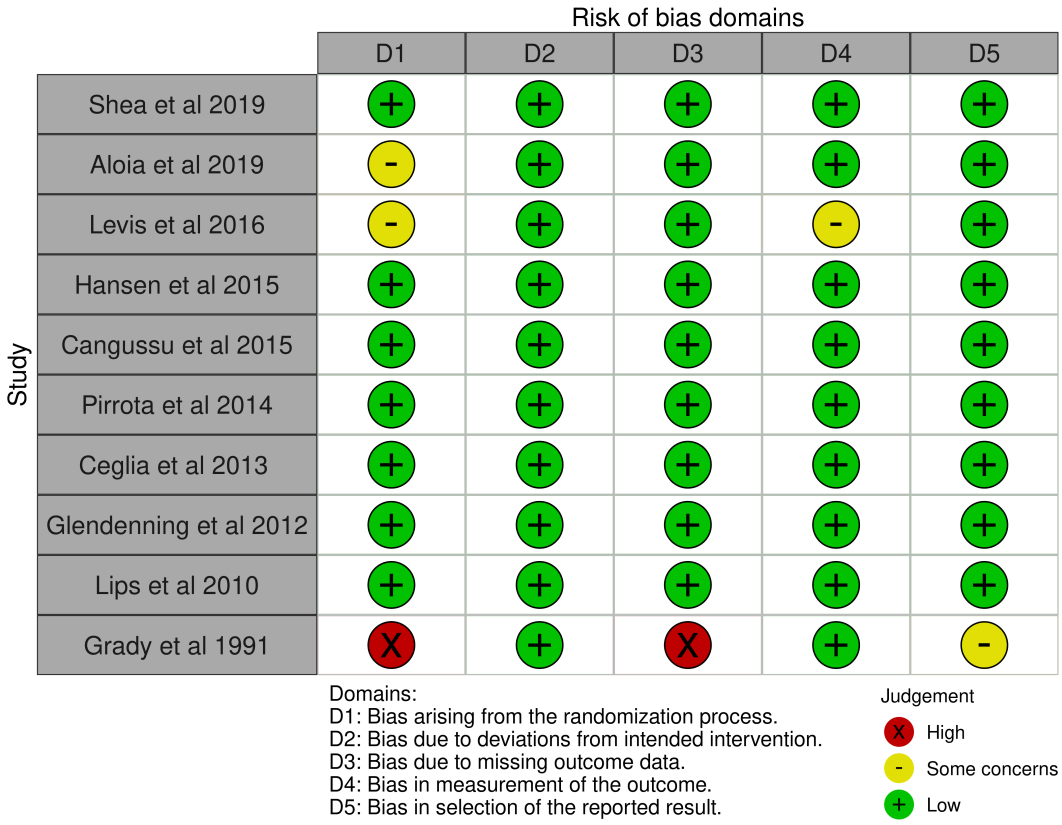


Figure S2. Effect of vitamin D supplementation on changes in serum vitamin D (25-hydroxyvitamin D [25(OH)D] and 1,25-dihydroxyvitamin D [1,25$(OH)_2$D]) levels compared to placebo.

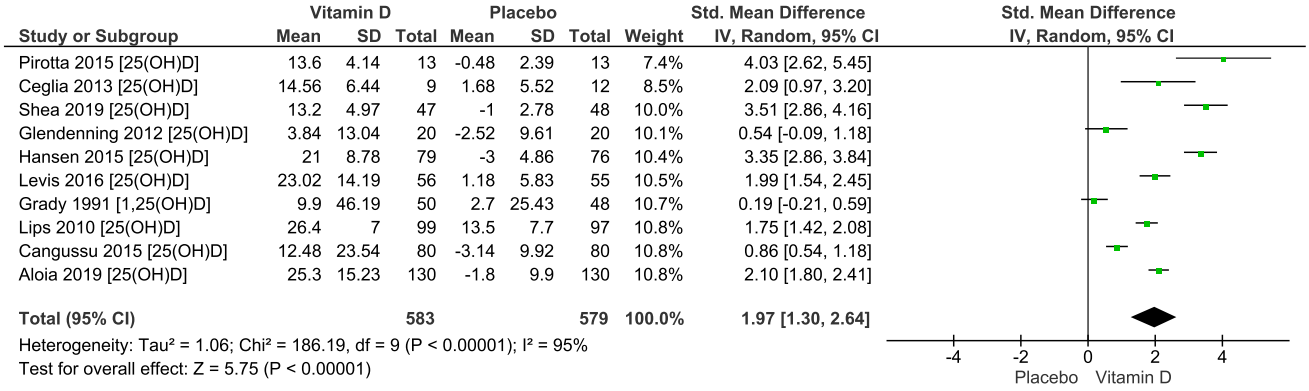
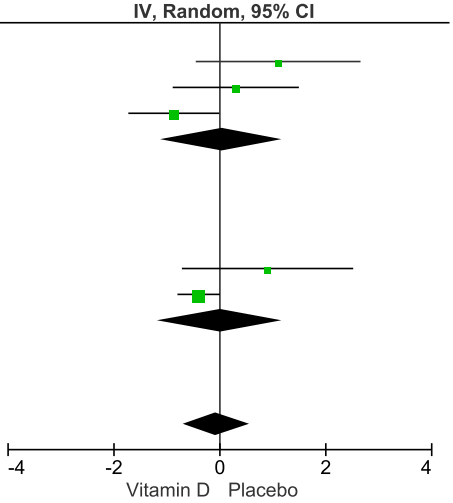


Figure S3. Subgroup analysis of handgrip strength changes in response to vitamin D supplementation based on **(A)** sex, vitamin D treatment **(B)** duration and **(C)** dose and **(D)** geographic origin of study, compared to placebo.

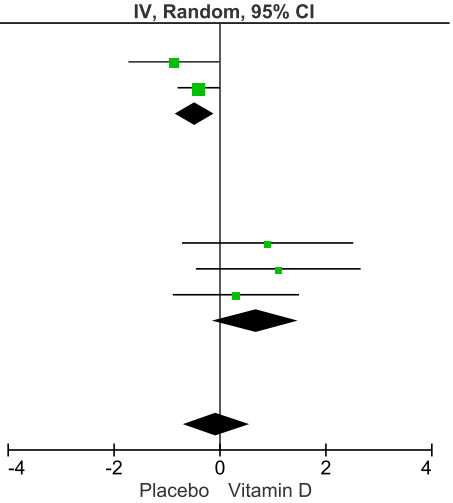
A

Study or Subgroup	Vitamin D Mean	SD	Total	Placebo Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Women and men								
Levis 2016	-1.65	5.74	57	-2.75	1.69	54	11.8%	1.10 [-0.46, 2.66]
Shea 2019	0.9	3.25	47	0.6	2.69	50	16.8%	0.30 [-0.89, 1.49]
Grady 1991	0.64	1.9	50	1.51	2.4	48	23.8%	-0.87 [-1.73, -0.01]
Subtotal (95% CI)			**154**			**153**	**52.4%**	**0.04 [-1.10, 1.19]**
Heterogeneity: Tau² = 0.66; Chi² = 5.69, df = 2 (P = 0.06); I² = 65%								
Test for overall effect: Z = 0.07 (P = 0.94)								
Women								
Cangussu 2015	0.6	4.81	80	-0.3	5.6	80	11.1%	0.90 [-0.72, 2.52]
Glendenning 2012	-1.05	2.49	311	-0.65	2.52	288	36.5%	-0.40 [-0.80, 0.00]
Subtotal (95% CI)			**391**			**368**	**47.6%**	**0.00 [-1.18, 1.18]**
Heterogeneity: Tau² = 0.48; Chi² = 2.34, df = 1 (P = 0.13); I² = 57%								
Test for overall effect: Z = 0.01 (P = 0.99)								
Total (95% CI)			**545**			**521**	**100.0%**	**-0.07 [-0.70, 0.55]**
Heterogeneity: Tau² = 0.24; Chi² = 8.13, df = 4 (P = 0.09); I² = 51%								
Test for overall effect: Z = 0.23 (P = 0.82)								
Test for subgroup differences: Chi² = 0.00, df = 1 (P = 0.96), I² = 0%								



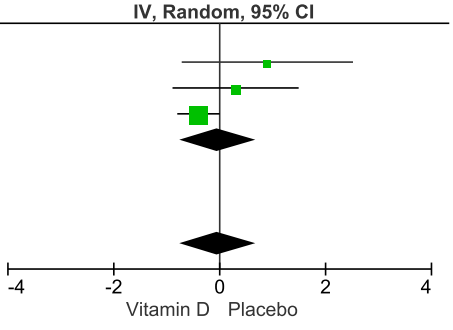
B

Study or Subgroup	Vitamin D Mean	SD	Total	Placebo Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Less than 6 months								
Grady 1991	0.64	1.9	50	1.51	2.4	48	23.8%	-0.87 [-1.73, -0.01]
Glendenning 2012	-1.05	2.49	311	-0.65	2.52	288	36.5%	-0.40 [-0.80, 0.00]
Subtotal (95% CI)			**361**			**336**	**60.3%**	**-0.48 [-0.85, -0.12]**
Heterogeneity: Tau² = 0.00; Chi² = 0.94, df = 1 (P = 0.33); I² = 0%								
Test for overall effect: Z = 2.61 (P = 0.009)								
More than 6 months								
Cangussu 2015	0.6	4.81	80	-0.3	5.6	80	11.1%	0.90 [-0.72, 2.52]
Levis 2016	-1.65	5.74	57	-2.75	1.69	54	11.8%	1.10 [-0.46, 2.66]
Shea 2019	0.9	3.25	47	0.6	2.69	50	16.8%	0.30 [-0.89, 1.49]
Subtotal (95% CI)			**184**			**185**	**39.7%**	**0.67 [-0.14, 1.49]**
Heterogeneity: Tau² = 0.00; Chi² = 0.74, df = 2 (P = 0.69); I² = 0%								
Test for overall effect: Z = 1.62 (P = 0.11)								
Total (95% CI)			**545**			**521**	**100.0%**	**-0.07 [-0.70, 0.55]**
Heterogeneity: Tau² = 0.24; Chi² = 8.13, df = 4 (P = 0.09); I² = 51%								
Test for overall effect: Z = 0.23 (P = 0.82)								
Test for subgroup differences: Chi² = 6.44, df = 1 (P = 0.01), I² = 84.5%								



C

Study or Subgroup	Vitamin D Mean	SD	Total	Placebo Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Moderate dose (853 - 1667 IU/d)								
Cangussu 2015	0.6	4.81	80	-0.3	5.6	80	15.3%	0.90 [-0.72, 2.52]
Shea 2019	0.9	3.25	47	0.6	2.69	50	24.1%	0.30 [-0.89, 1.49]
Glendenning 2012	-1.05	2.49	311	-0.65	2.52	288	60.6%	-0.40 [-0.80, 0.00]
Subtotal (95% CI)			**438**			**418**	**100.0%**	**-0.03 [-0.74, 0.68]**
Heterogeneity: Tau² = 0.17; Chi² = 3.29, df = 2 (P = 0.19); I² = 39%								
Test for overall effect: Z = 0.09 (P = 0.93)								
Total (95% CI)			**438**			**418**	**100.0%**	**-0.03 [-0.74, 0.68]**
Heterogeneity: Tau² = 0.17; Chi² = 3.29, df = 2 (P = 0.19); I² = 39%								
Test for overall effect: Z = 0.09 (P = 0.93)								
Test for subgroup differences: Not applicable								



D

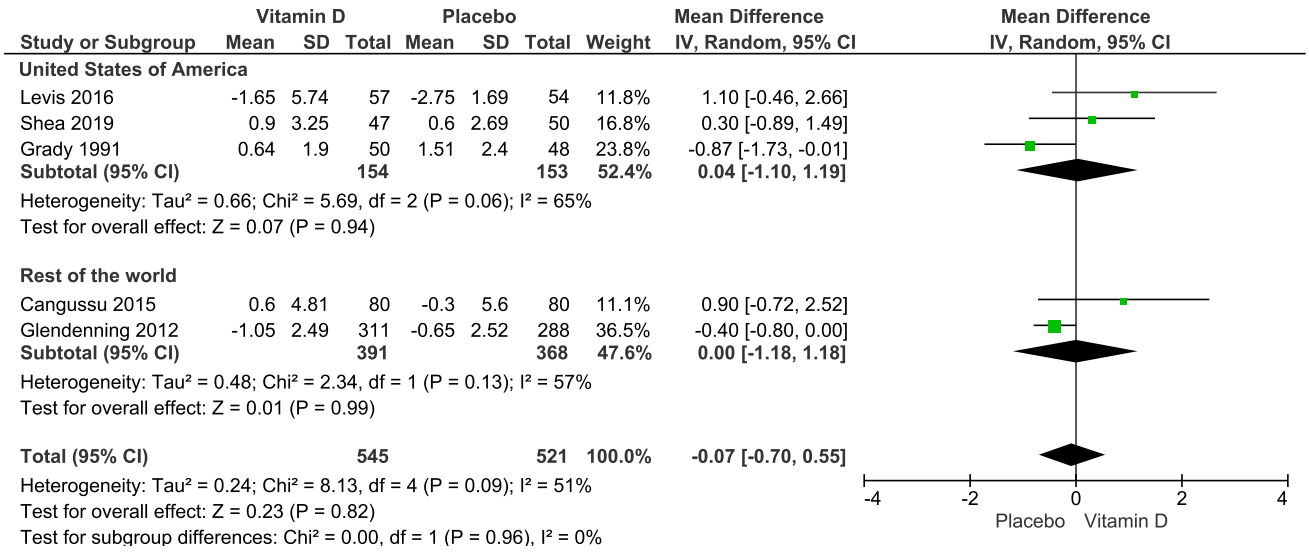
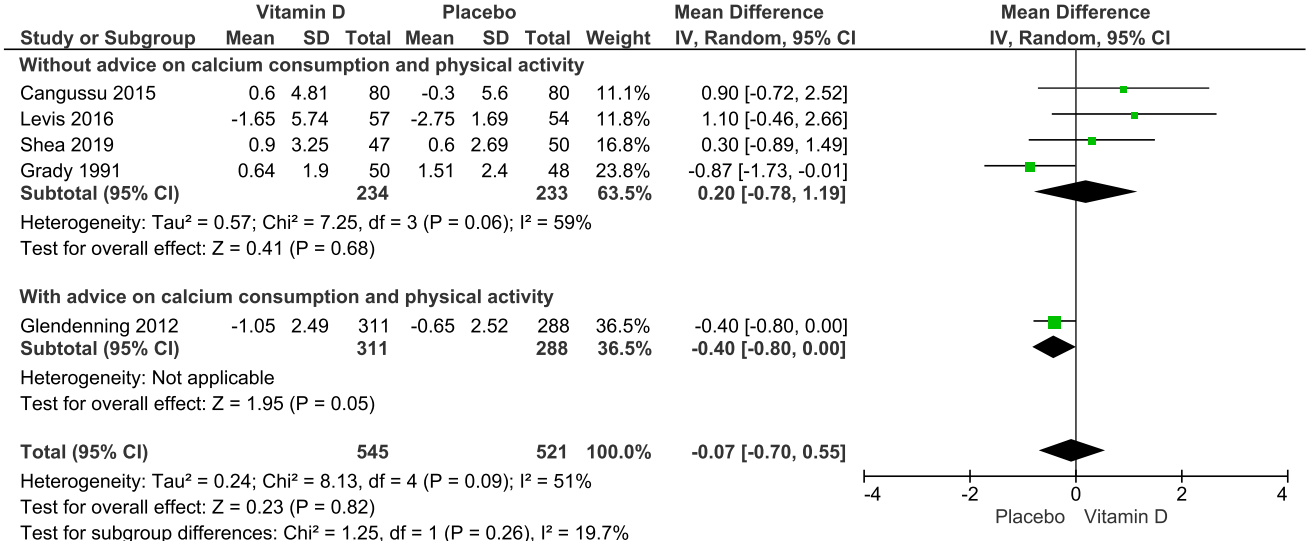

Vitamin D
Placebo
Mean Difference
Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
United States of America
Levis 2016 -1.65 5.74 57 -2.75 1.69 54 11.8% 1.10 [-0.46, 2.66]
Shea 2019 0.9 3.25 47 0.6 2.69 50 16.8% 0.30 [-0.89, 1.49]
Grady 1991 0.64 1.9 50 1.51 2.4 48 23.8% -0.87 [-1.73, -0.01]
Subtotal (95% CI) 154 153 52.4% 0.04 [-1.10, 1.19]
Heterogeneity: Tau² = 0.66; Chi² = 5.69, df = 2 (P = 0.06); I² = 65%
Test for overall effect: Z = 0.07 (P = 0.94)
Rest of the world
Cangussu 2015 0.6 4.81 80 -0.3 5.6 80 11.1% 0.90 [-0.72, 2.52]
Glendenning 2012 -1.05 2.49 311 -0.65 2.52 288 36.5% -0.40 [-0.80, 0.00]
Subtotal (95% CI) 391 368 47.6% 0.00 [-1.18, 1.18]
Heterogeneity: Tau² = 0.48; Chi² = 2.34, df = 1 (P = 0.13); I² = 57%
Test for overall effect: Z = 0.01 (P = 0.99)
Total (95% CI) 545 521 100.0% -0.07 [-0.70, 0.55]
Heterogeneity: Tau² = 0.24; Chi² = 8.13, df = 4 (P = 0.09); I² = 51%
Test for overall effect: Z = 0.23 (P = 0.82)
Test for subgroup differences: Chi² = 0.00, df = 1 (P = 0.96), I² = 0%
-4
-2
0
2
4
Placebo
Vitamin D

Figure S4. Sensitivity analysis based on the effect of lifestyle factors on changes in **(A)** handgrip strength **(B)** serum vitamin D (25-hydroxyvitamin D [25(OH)D] and 1,25-dihydroxyvitamin D [1,25$(OH)_2$D]), in response to vitamin D supplementation compared to placebo.

A



B

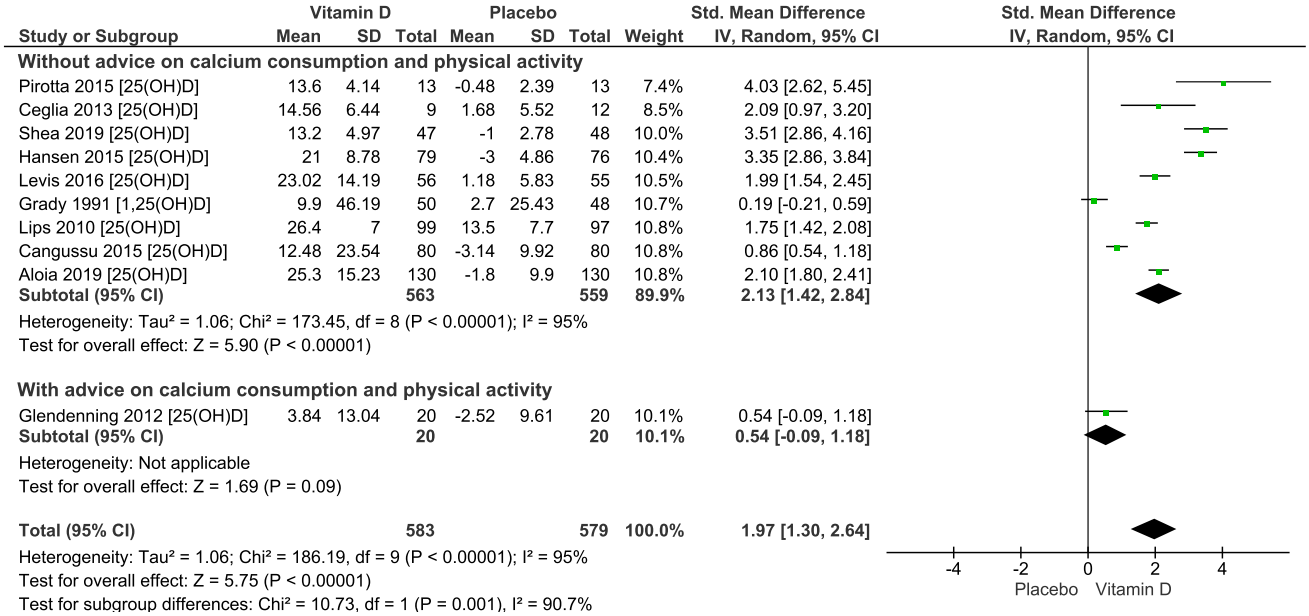
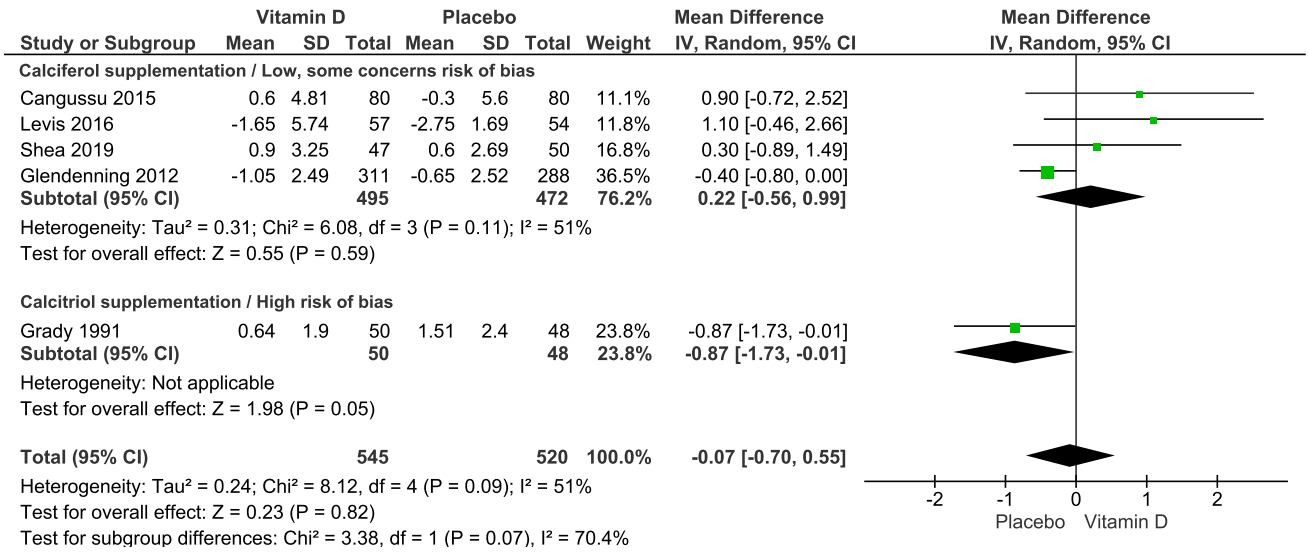
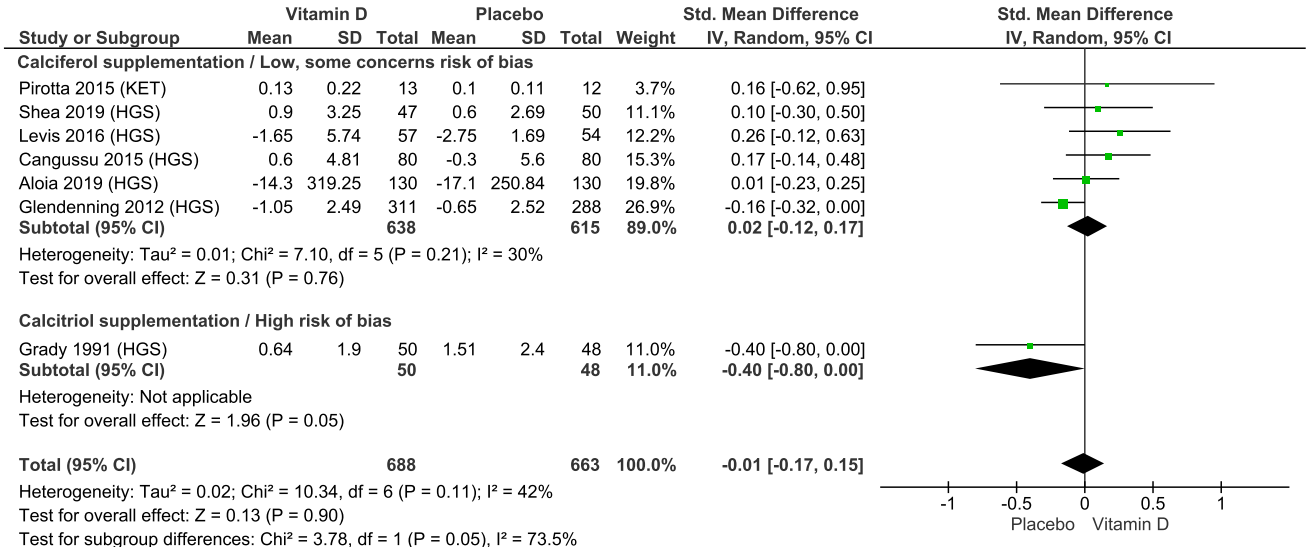


Figure S5. Sensitivity analysis based on the effect of type of vitamin D supplementation or risk of bias on changes in **(A)** handgrip strength (HGS), **(B)** general muscle strength (HGS and knee extension test at 180 degrees (KET)) and **(C)** serum vitamin D (25-hydroxyvitamin D [25(OH)D] and 1,25-dihydroxyvitamin D [1,25(OH)$_2$D]) levels, in response to vitamin D supplementation compared to placebo.

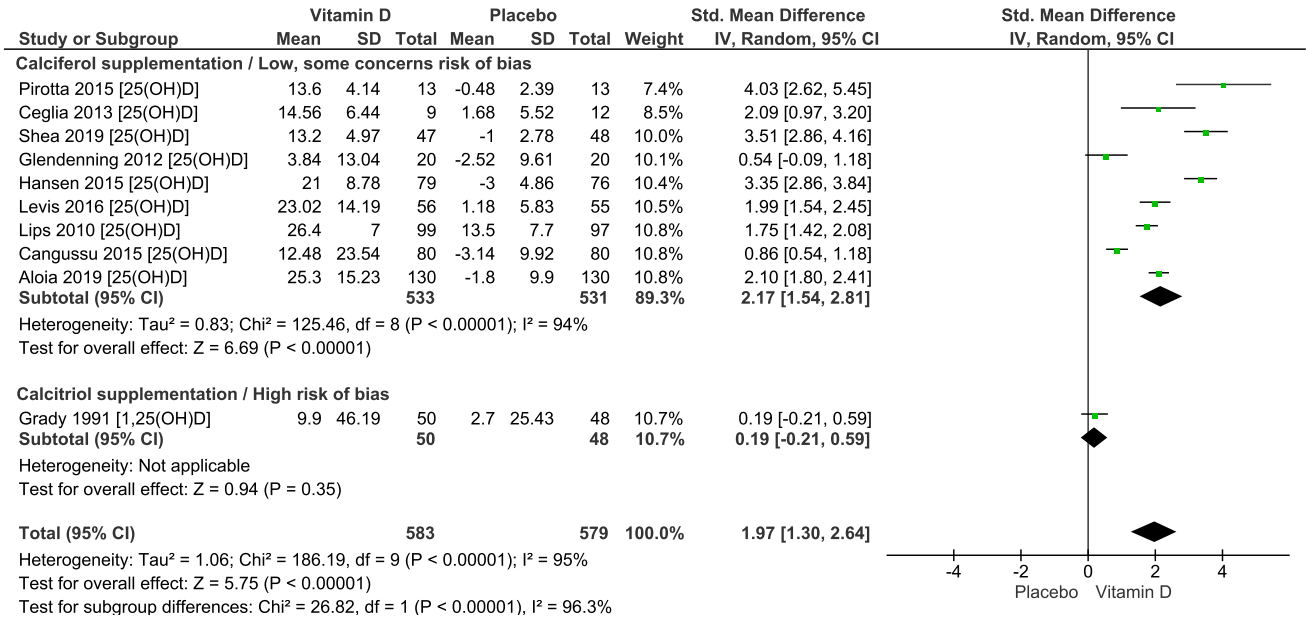
A



B



C


Vitamin D
Placebo
Std. Mean Difference
Std. Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
Calciferol supplementation / Low, some concerns risk of bias
Pirotta 2015 [25(OH)D] 13.6 4.14 13 -0.48 2.39 13 7.4% 4.03 [2.62, 5.45]
Ceglia 2013 [25(OH)D] 14.56 6.44 9 1.68 5.52 12 8.5% 2.09 [0.97, 3.20]
Shea 2019 [25(OH)D] 13.2 4.97 47 -1 2.78 48 10.0% 3.51 [2.86, 4.16]
Glendenning 2012 [25(OH)D] 3.84 13.04 20 -2.52 9.61 20 10.1% 0.54 [-0.09, 1.18]
Hansen 2015 [25(OH)D] 21 8.78 79 -3 4.86 76 10.4% 3.35 [2.86, 3.84]
Levis 2016 [25(OH)D] 23.02 14.19 56 1.18 5.83 55 10.5% 1.99 [1.54, 2.45]
Lips 2010 [25(OH)D] 26.4 7 99 13.5 7.7 97 10.8% 1.75 [1.42, 2.08]
Cangussu 2015 [25(OH)D] 12.48 23.54 80 -3.14 9.92 80 10.8% 0.86 [0.54, 1.18]
Aloia 2019 [25(OH)D] 25.3 15.23 130 -1.8 9.9 130 10.8% 2.10 [1.80, 2.41]
Subtotal (95% CI) 533 531 89.3% 2.17 [1.54, 2.81]
Heterogeneity: Tau² = 0.83; Chi² = 125.46, df = 8 (P < 0.00001); I² = 94%
Test for overall effect: Z = 6.69 (P < 0.00001)
Calcitriol supplementation / High risk of bias
Grady 1991 [1,25(OH)D] 9.9 46.19 50 2.7 25.43 48 10.7% 0.19 [-0.21, 0.59]
Subtotal (95% CI) 50 48 10.7% 0.19 [-0.21, 0.59]
Heterogeneity: Not applicable
Test for overall effect: Z = 0.94 (P = 0.35)
Total (95% CI) 583 579 100.0% 1.97 [1.30, 2.64]
Heterogeneity: Tau² = 1.06; Chi² = 186.19, df = 9 (P < 0.00001); I² = 95%
Test for overall effect: Z = 5.75 (P < 0.00001)
Test for subgroup differences: Chi² = 26.82, df = 1 (P < 0.00001), I² = 96.3%
-4
-2
0
2
4
Placebo
Vitamin D